

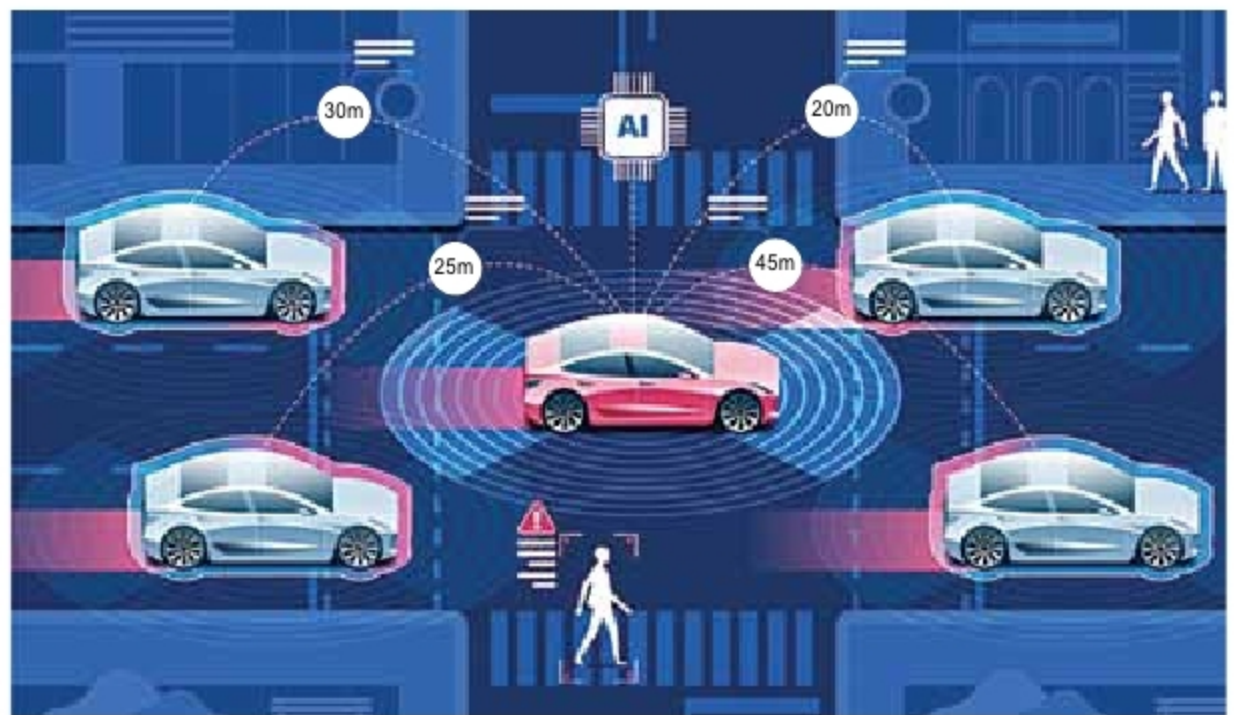
their efficiency, availability, lack of wear elements, and regenerative braking capability. Several other concepts are also being developed, such as asymmetric motors, e-propulsion systems, high efficiency switched reluctance motors and inverter-driven AC induction motors. Another emerging technology is axial flux motors, where magnetic flux is produced along the axis instead of the sideways of the rotor.

The most favourite technology being used for applications like scooters and e-bikes is electric direct-drive in-wheel motors or hub motors. These provide distinct benefits like individual control of each drive wheel's speed, torque, traction, and regenerative braking. Fully hidden hub motors take up hardly any additional space.

The problems faced in using the hub motors are unsprung weight and inertia. The wheel with the motor in its hub becomes heavier, making it harder for the vehicle to accelerate. Also, when the wheel hits a bump, the suspension is required to make extra effort to put back the wheel on to the ground, making for a rough ride. With hub motors, the rotational mass of the wheel increases, and greater power is required to move the vehicle.

These problems are surmounted by using more active suspensions that forestall bumps to prevent a rough ride. With each wheel having one motor, motors become smaller, and weight addition is considerably reduced by having an appropriate housing and wheel structure.

**Solid-state batteries.** The conventional lead-acid batteries are large in size, heavy, and charge slowly. Lithium-ion batteries solve most of these issues as these are smaller in size, lighter in weight for the same voltage,



LiDAR technology uses reflected light to measure distances (Credit: <https://www.azosensors.com>)

and have more charge cycles, thereby lasting longer. These cost more than lead-acid batteries but are safer for the environment as these hardly emit any carbon.

For an EV having the same range as a conventional vehicle, the weight of lithium-ion batteries required would be considerably more than a tank of gas.

**Solid-state battery technology** provides batteries that are as small as a gas tank, weigh a similar amount, get fully charged in only a few minutes, and provide an energy density 2.5 times of lithium-ion batteries. The added advantage is that these are not flammable. However, these batteries are expensive, and there are concerns about low-temperature performance and mechanical durability.

**Faster electric charging.** Rapid advancements in electric chargers and charging infrastructure is taking place to cut down the charging time drastically. Gbatteries has recently developed an electric car charger that lets the car charge quickly. Another emerging technology is wireless or induction charging that is similar to wireless mobile phone charging but at considerably higher power.

Qualcomm Halo has been working on a 22kW wireless charger with the possibility of charging vehicles while they drive up to 113kmph. Nissan, Mercedes Benz, and BMW are also developing wireless charging capability in their vehicles.

## Conclusion

Today, the automotive industry is one of the most data-driven industries. The modern smart and intelligent vehicles use software systems and collect enormous data pertaining to the functioning of various systems, drivers' behaviour, preferences and condition, routes and destinations, traffic patterns, and maintenance. Therefore data security and protection are of utmost importance and the most challenging aspects.

There is a pressing need to secure and protect the collected data to preclude the hackers from forcibly taking control over various vehicle functions and capturing the vehicle by manipulating the data. With the continuous increase in demand, more and more companies are bringing out electric vehicles, which are contributing a great deal in reducing carbon emissions and global warming.

IT and AI are playing a major role in the customisation of vehicles and alerting the owners when their vehicles require maintenance and need replacement of certain components. The development of self-driving vehicles with increasing safety is fast progressing and may become a reality in the near future. Technological advancements are constantly revolutionising not just the automotive industry but human transportation in general. The future smart car will make people travel with greater safety, provide them with more useful information and let them just enjoy the ride like never before. **EFY**



It is predicted that by 2030 you will be able to hail a driverless taxi (Credit: <https://www.pymnts.com>)